



COSC322

Introduction to Artificial Intelligence – Lab04/05

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Requirements

COSC 322 Lab 02 (Warmup-2)

Turn this into a GUI-based Spectator by initialize an instance of `BaseGameGUI.java` and assign it to `this.gamegui`, and complete the following

- Replace the code in your method `onLogin()` with the following two statements:

```
userName = gameClient.getUserName();
if(gamegui != null) {
    gamegui.setRoomInformation(gameClient.getRoomList());
}
```

- Add Implementation in the method `handleGameMessage(...)` to handle the two types game messages:
 1. `GameMessage.GAME_STATE_BOARD` (call `BaseGameGui.setGameState(...)` to set the game board)
 2. `GameMessage.GAME_ACTION_MOVE` (call `BaseGameGui.updateGaemState(...)` to update the game board)
 - In a game player, upon receiving this message about your opponent's move, you will also need to calculate your move and send your move to the server using the method `GameClient.sendMessage(...)` (these are the core tasks of this project you will have to by the middle of March)

For a detailed description of the message format, see API document on `GamePlayer.handleGameMessage(...)`

If everything goes well, your program can be used as a spectator. To test your implementation, start two instances of `HumanPlayer` and have them join a game room, play a few moves, and then start your program and join the same room.

Packages

GAME_STATE_BOARD
GAME_ACTION_START
- PLAYER_BLACK
- PLAYER_WHITE
GAME_ACTION_MOVE

Hash Key
AmazonsGameMessage.QUEEN_POS_CURR
AmazonsGameMessage.QUEEN_POS_NEXT
AmazonsGameMessage.ARROW_POS

COSC322Test.javaGameMessage.classBaseGameGUI.classAmazonsG

Decompiled .class file, bytecode version: 52.0 (Java 8)

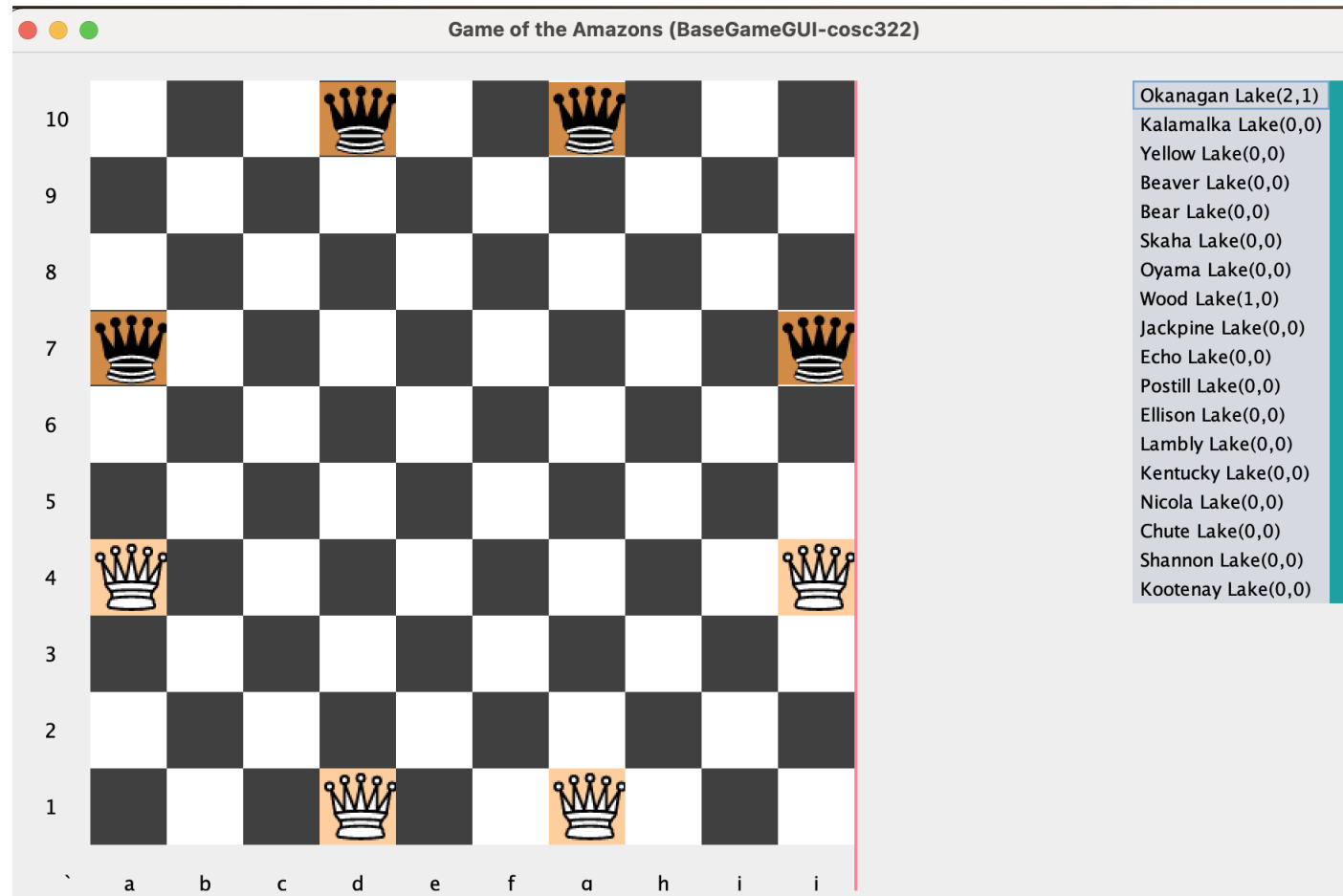
```
6 package ygraph.ai.smartfox.games;
7
8 import com.smartfoxserver.v2.entities.data.ISFSObject;
9 import com.smartfoxserver.v2.entities.data.SFSObject;
10 import sfs2x.client.requests.ExtensionRequest;
11
12 4 usages 1 inheritor
12 @ public class GameMessage {
13     1 usage
13     public static final String GAME_ACTION_START = "cosc322.game-action.start";
14     1 usage
14     public static final String GAME_ACTION_MOVE = "cosc322.game-action.move";
15     no usages
15     public static final String GAME_STATE_JOIN = "cosc322.game-state.join";
16     1 usage
16     public static final String GAME_STATE_BOARD = "cosc322.game-state.board";
17     no usages
17     public static final String GAME_STATE_PLAYER_LOST = "cosc322.game-state.player-lost";
18     no usages
18     public static final String GAME_TEXT_MESSAGE = "cosc322.game-state.text-message";
19     protected ExtensionRequest req = null;
20     protected String messageType = null;
21     protected ISFSObject params = null;
22
23     no usages
23     public GameMessage(String messageType) {
24         this.messageType = messageType;
25         this.params = new SFSObject();
26     }
27 }
```

COSC322Test.javaMap.javaAmazonsGameMessage.classGameMessage.classBaseGame

Decompiled .class file, bytecode version: 52.0 (Java 8)

```
8 > import ...
11 5 usages
12 public class AmazonsGameMessage extends GameMessage {
13     2 usages
13     public static final String QUEEN_POS_CURR = "queen-position-current";
14     no usages
14     public static final String QUEEN_POS_NEXT = "queen-position-next";
15     no usages
15     public static final String ARROW_POS = "arrow-position";
16     no usages
16     public static final String GAME_STATE = "game-state";
17     1 usage
17     public static final String PLAYER_BLACK = "player-black";
18     1 usage
18     public static final String PLAYER_WHITE = "player-white";
19     no usages
19     private boolean msgCompiled = false;
20
21     no usages
21     public AmazonsGameMessage(String msgType) { super("cosc322.game-action.move"); }
22
23     no usages
23     public void setMessageDetails(String key, Object value) {
24         if (super.messageType.equals("cosc322.game-action.move")) {
25             this.params.putIntArray(key, (Collection) value);
26         } else {
27             this.params.putUtfString(key, (String) value);
28         }
29     }
30
31     public void compile() {
32         super.req = new ExtensionRequest(this.messageType, this.params);
33         this.msgCompiled = true;
34     }
35 }
```

Results



Bugs

- ♦ If white-queens don't show up

- Mac & Win:   ygraph-ai-smartfox-client-2.1.jar  

- Download

- /Users/user_name/.m2/repository/ubc-yong-gao/ygraph-ai-smartfox-client/2.1/ygraph-ai-smartfox-client-2.1.jar!/ygraph/ai/smartfox/games/amazons/images/white-queen.png